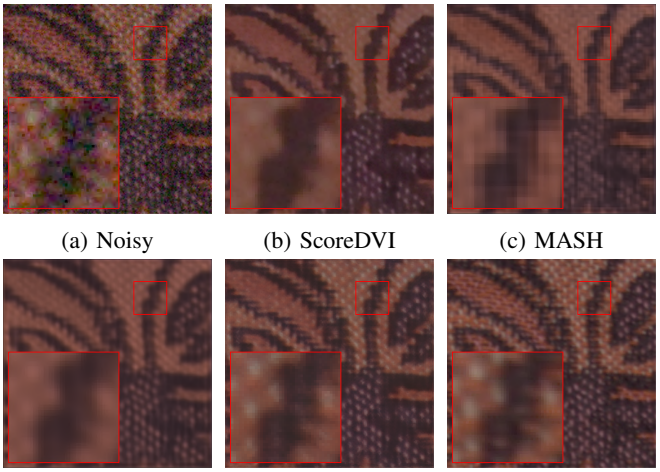
# Explicit Multi-Scale Blind-Spot Denoising with Edge-Aware Spatial Loss

Ming Cheng[1] and Qingxuan Shi[2] and Fang Yang[3] and Wanzhi Hong[4]

***Abstract*— Blind-spot denoising (BSD) methods have made significant progress in self-supervised image denoising; however, noise in real-world images exhibits substantial heterogeneity across both spatial and scale dimensions. Existing single-scale feature modeling and spatially uniform loss functions often lead to over-smoothing in textured areas and insufficient denoising in smooth regions, struggling to balance detail preservation with noise suppression. To address these challenges, this paper proposes an Explicit Multi-Scale Blind-Spot Denoising (EMBSD) framework. Architecturally, we construct a four-scale multi-resolution network by deploying independent Dilated Blind-Spot Networks at the original resolution, as well as at 1/2, 1/4, and 1/8 scales, simultaneously capturing local fine-grained textures and large-scale noise fluctuations. For optimization, we introduce an edge-aware spatial loss weighting strategy that assigns different reconstruction loss weights to edge and flat regions based on gradient magnitude, thereby achieving spatially differentiated denoising. Experiments on real-world image denoising datasets demonstrate that EMBSD achieves competitive or superior performance across different datasets, with consistent improvements in structural detail preservation and noise suppression.**



(a) Noisy (b) ScoreDVI (c) MASH

(d) BSN-INS (e) Ours (f) Clean

Fig. 1: Visual comparison of zero-shot denoising methods on real-world image.

## I. Introduction

Image denoising, which aims to restore a clean image from its noisy observation, constitutes a fundamental yet challenging low-level vision task in computer vision. With the rapid advancement of deep learning, supervised denoising methods [1], [2] have become mainstream and achieved remarkable performance. However, these approaches heavily rely on large-scale paired noisy-clean datasets, incurring high acquisition costs and facing practical difficulties in obtaining ideal reference images.

Recently, self-supervised image denoising techniques [3]–[12] have emerged, enabling models to be trained directly from noisy observations without requiring clean ground-truth images, thereby opening new research directions. Among these, blind-spot networks (BSNs) have established themselves as a representative framework. By excluding the target pixel itself during prediction, BSNs avoid learning an identity mapping and allow training without clean references.

Nevertheless, real-world noisy images exhibit significant spatial-scale heterogeneity, posing a dual challenge to existing BSNs. From a scale perspective, the interaction between noise and image structure varies with spatial frequency: in high-frequency textured regions, noise becomes intricately entangled with fine details and local variations, making it difficult to distinguish from genuine structures; conversely, in low-frequency homogeneous areas, noise manifests as large-scale fluctuations that disrupt region consistency. Most existing BSNs rely predominantly on single-scale feature modeling. While some methods, such as APBSN [10], have implicitly encoded multi-scale information via mechanisms like pixel shuffling, they still lack explicit decomposition capabilities to comprehensively capture noise patterns across different scales.

From a spatial perspective, the structural complexity of noisy images varies significantly across different regions. Conventional BSNs apply a uniform loss function to all pixels during training. This spatially homogeneous optimization objective tends to cause over-smoothing in textured areas while offering insufficient denoising flexibility in smooth regions, thereby struggling to balance the inherent trade-off between detail preservation and noise suppression.

To address these challenges, this paper proposes an Explicit Multi-Scale Blind-Spot Denoising (EMBSD) framework, that synergistically combines architectural design and optimization strategy. Specifically, we construct a four-scale network by deploying independent dilated blind-spot network (DBSN) [13] at the original resolution as well as at 1/2, 1/4, and 1/8 scales. High-resolution branches are dedicated to capturing local textures and fine structures, while low-resolution branches model long-range contextual dependencies and large-scale noise fluctuations.

To further leverage the representational power of the multi-scale architecture, we introduce an edge-aware spatial loss weighting strategy. Gradient magnitude is first employed to measure structural complexity, partitioning pixels into edge and flat regions, to which different loss weights are assigned accordingly. In edge regions, higher weights encourage the model to adhere to targets, thereby preserving structural

details; in flat regions, lower weights grant greater flexibility for noise suppression.

By addressing heterogeneity across both scale and spatial dimensions, this joint design of architecture and loss function helps EMBSD better separate complex noise–structure interactions. As illustrated in Fig. 1, this synergy leads to superior visual results compared to existing zero-shot denoisers, demonstrating improved structure preservation and noise suppression.

The main contributions of this paper are summarized as follows:

- We propose an Explicit Multi-Scale Blind-Spot Denoising (EMBSD) framework. Unlike existing single-scale blind-spot networks, it leverages explicit multi-resolution branches to simultaneously capture fine-grained textures and macroscopic noise fluctuations, thereby enhancing cross-scale feature modeling capabilities.
- We design an edge-aware spatial loss weighting strategy. This approach moves beyond conventional uniform loss functions by adaptively assigning weights to achieve spatially differentiated denoising, preserving structural details while improving denoising effectiveness in flat regions.
- Through the deep integration of architecture and loss formulation, EMBSD outperforms existing zero-shot denoising methods on real-world datasets, demonstrating significantly enhanced structure preservation and noise suppression.

## II. Related Work

### A. Supervised Denoising

Supervised image denoising aims to learn a direct mapping from noisy to clean images from paired datasets. Early CNN-based methods achieved substantial improvements over traditional approaches. Subsequent works, including FFDNet [1] and CBDNet [2] focused on enhancing network architectures and refining noise modeling to further improve performance. Despite their impressive results, supervised approaches require large-scale, carefully aligned noisy-clean or noise-noise image pairs, which are expensive and labor-intensive to acquire in practical scenarios. Moreover, models trained on synthetic noise often struggle to generalize to real-world noise distributions, limiting their practical applicability.

### B. Self-supervised Denoising

Self-supervised denoising methods eliminate the dependence on clean images by exploiting only noisy inputs during training. The prominent paradigm is BSNs, which prevents the networks from accessing the target pixel during prediction to avoid trivial identity mapping. Early works, including Noise2Void [3] and Noise2Self [4], demonstrated the effectiveness of this strategy. Extensions such as Neighbor2Neighbor [5] improved blind-spot design and masking strategies. Other methods, including R2R [6], generate synthetic noisy pairs from noisy inputs, though these approaches require prior knowledge of the noise distribution. Techniques like pixel-shuffle downsampling and asymmetric blind-spot strategies, as adopted in LG-BPN [7], reduce local noise correlations and enhance texture recovery. More recently, transformer-based blind-spot models such as SelfFormer [8] have leveraged directional self-attention and global context to further improve denoising performance. Despite these advances, self-supervised approaches remain challenged by highly correlated noise and often require careful network or masking design to achieve optimal results.

### C. Zero-shot Denoising

Zero-shot denoising methods are applied directly to a test image without relying on external datasets. The pioneering work DIP [14] demonstrated that the inductive biases of CNNs can capture natural image statistics more readily than noise, enabling effective single-image denoising. Building on this idea, Self2Self [15] combines blind-spot mechanisms with dropout ensembling, Noise2Fast [16] uses checkerboard downsampling to generate training pairs, and ZS-N2N [17] introduces refined downsampling strategies for single-image adaptation. Pixel2Pixel [18] exploits non-local self-similarity within the input image to synthesize abundant training samples, while NAC [19] creates auxiliary noisy pairs by adding synthetic noise. To handle spatially correlated noise, ScoreDVI [20] incorporates score priors into Bayesian inference, and MASH [21] applies adaptive masking along with local pixel shuffling to mitigate noise correlation. Overall, zero-shot approaches offer great flexibility and strong robustness in dealing with unseen noise and real-world scenarios, avoiding the dependency on large pre-collected datasets inherent in supervised and self-supervised methods.

The proposed EMBSD belongs to the category of zero-shot self-supervised Blind-spot denoising (BSD) methods. However, unlike existing BSN-based approaches that typically model the denoising process at a single scale and employ spatially uniform loss functions, EMBSD explicitly models multi-scale noise characteristics through a multi-resolution blind-spot architecture. Furthermore, an edge-aware loss weighting strategy is introduced to enhance spatial adaptivity, enabling the network to better preserve structural details while suppressing noise.

## III. Method

In this section, we introduce the proposed Explicit Multi-Scale Blind-Spot Denoising (EMBSD) framework. We first describe the blind-spot training scheme with pixel refilling, then present the explicit multi-scale blind-spot architecture, and finally introduce the edge-aware spatial loss weighting strategy. Overview of the proposed framework is illustrated in Fig. 2.

### A. Blind-Spot Network with Pixel Refilling

As the foundation of our framework, BSNs are self-supervised denoising models that prevent the network from accessing the target pixel during training, forcing predictions to rely solely on surrounding context. In this work, we adopt

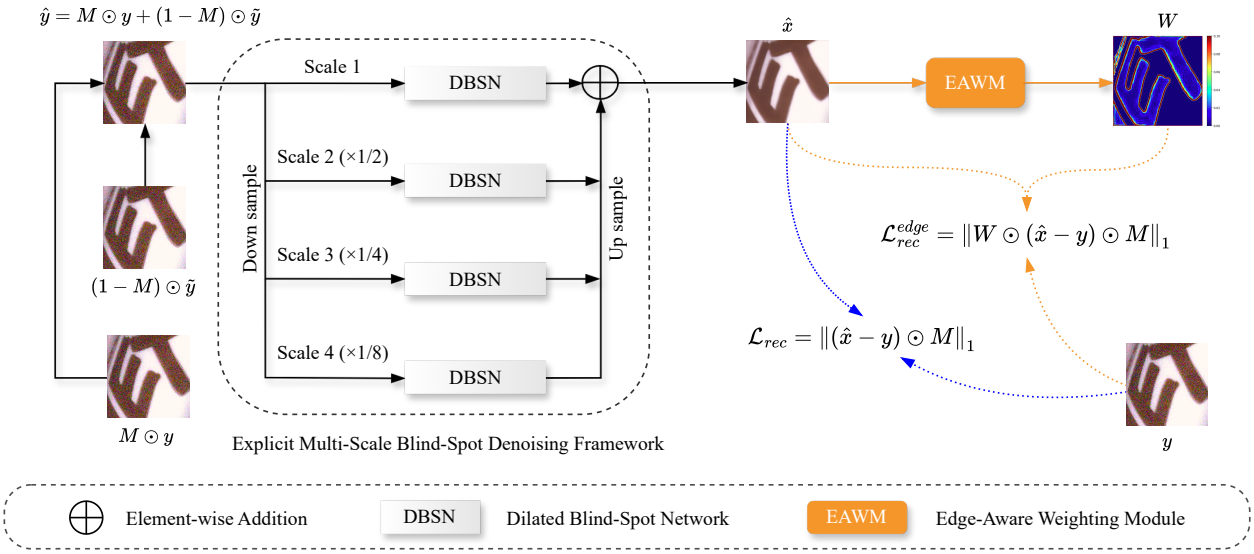



Fig. 2: Overview of the proposed EMBSD framework. The framework explicitly captures multi-scale noise characteristics via four resolution branches, each processed by an independent DBSN. Outputs from lower-resolution branches are upsampled and aggregated with the original-scale prediction via element-wise addition. The training involves two stages: the first utilizes a base denoising loss (blue dashed lines), while the second is guided by the Edge-Aware Weighting Module (EAWM) using an edge-aware loss (orange dashed lines) to suppress noise while preserving structural details.

the DBSN [13] as the fundamental building block for each scale branch. Following the pixel-shuffling strategy [22], an auxiliary image $\tilde{y}$ is constructed from the original noisy observation $y$. In each training iteration, a binary mask $M \in \{0,1\}^{H \times W}$ is independently sampled from a Bernoulli distribution. The composite input $\hat{y}$ is then formed by combining the unmasked original pixels with the auxiliary pixels at the masked positions:

$$\hat{y} = M \odot y + (1 - M) \odot \tilde{y}, \tag{1}$$

where $\odot$ denotes element-wise multiplication. This combined image $\hat{y}$ serves as the common input to all scale branches in our multi-scale framework. The blind-spot constraint is enforced by training the network to predict the original noisy pixels at masked locations, with the loss defined on the fused multi-scale output (detailed in Sec III-C).

### B. Explicit Multi-Scale Blind-Spot Architecture

Conventional blind-spot networks operate at a single spatial resolution, which limits their ability to capture noise patterns that exhibit different characteristics across scales. To overcome this limitation, we propose an explicit multi-scale blind-spot architecture with four independent branches, where high-resolution branches capture local textures and low-resolution branches model long-range dependencies, integrating local detail with global context.

Given the combined input image $\hat{y}$ in (1), we first generate its downsampled versions $\hat{y}_{1/2}$, $\hat{y}_{1/4}$, and $\hat{y}_{1/8}$ using bilinear interpolation. For each scale $s \in \{1, 1/2, 1/4, 1/8\}$, we employ an independent DBSN branch $\mathcal{F}_{DBSN}^{(s)}$ with identical architecture but separate learnable parameters. The detailed architecture of the DBSN is illustrated in Fig. 3. Each branch produces a denoised estimate at its corresponding scale:

$$\hat{x}_s = \mathcal{F}_{DBSN}^{(s)}(\hat{y}_s), \quad s \in \{1, 1/2, 1/4, 1/8\}, \tag{2}$$

where $\hat{y}_1 = \hat{y}$ denotes the original resolution input. The feature maps from the lower-resolution branches ($s = 1/2, 1/4, 1/8$) are then upsampled to the original resolution via bilinear interpolation:

$$\hat{x}_s^{\uparrow} = \mathrm{Up}(\hat{x}_s), \quad \text{for } s \neq 1. \tag{3}$$

Finally, the multi-scale predictions are aggregated through element-wise summation to yield the final denoised image:

$$\hat{x} = \sum_s \hat{x}_s^{\uparrow}, \tag{4}$$

where $\hat{x}_1^{\uparrow} = \hat{x}_1$ for the original scale. This multi-scale design allows more comprehensive contextual modeling than conventional single-scale BSNs. By combining information across scales, the network can better disentangle complex noise patterns and preserve structural details.

### C. Edge-Aware Spatial Loss Weighting

Standard blind-spot training employs a uniform reconstruction loss over all pixels, treating flat and textured regions equally. This uniform objective often leads to over-smoothing of edges or residual noise in homogeneous areas. Moreover, in a multi-scale setting, different branches may contribute unevenly during optimization, causing certain scales to dominate while others remain under-optimized. To address these issues, we introduce a spatially adaptive weighting mechanism guided by local image structures.

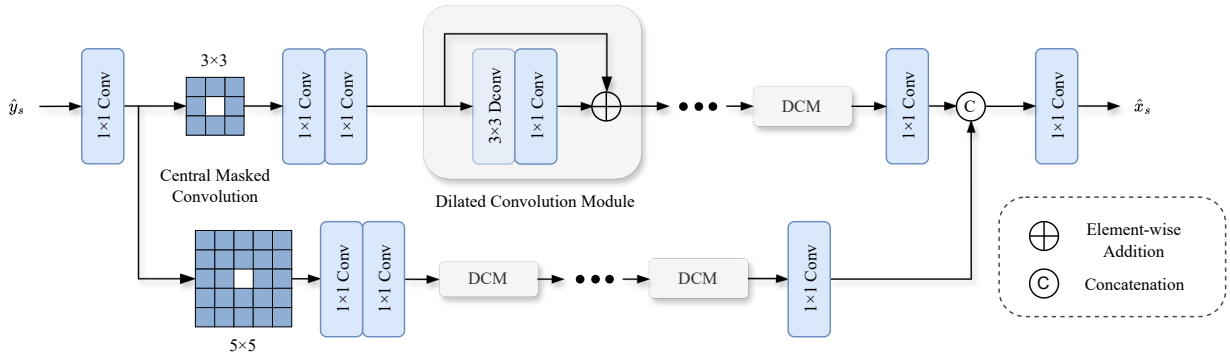



Fig. 3: Visualization of the DBSN Architecture. The architecture is derived from the 3×3 and 5×5 central masked convolutional designs introduced in [10], each Dilated Convolution (DC) module consists of a single 3×3 dilated convolution with a stride s. Each convolutional layer employs 64 channels by default. We stack four DC modules along each path of the network.

Specifically, the training process is divided into two stages: an initial training stage and an edge-aware optimization stage. First, a stabilized intermediate denoised estimate is obtained through the initial training stage, and a weight map that identifies edge regions and flat regions is generated based on gradient magnitude. During the edge-aware optimization stage, this weight map serves as a fixed spatial prior to weight both the reconstruction loss and the multi-scale consistency loss. This design enhances the focus on edge structures while simultaneously suppressing noise in flat regions.

**Loss formulation.** In the initial training stage, the edge structure estimated from the intermediate denoised image is not yet reliable, therefore, no spatial weighting is applied. A uniform reconstruction loss is employed for all pixels. Adopting the blind-spot training scheme described in Sec III-A, we denote the binary mask sampled each iteration as $M \in \{0,1\}^{H\times W}$. Let $\hat{\mathbf{x}}$ be the fused multi-scale output from (4) and $\hat{\mathbf{x}}_{base}$ be the output from the original-resolution branch. The loss is computed only on masked pixels, i.e., where $M_{i,j} = 1$. The initial training objective is formulated as:

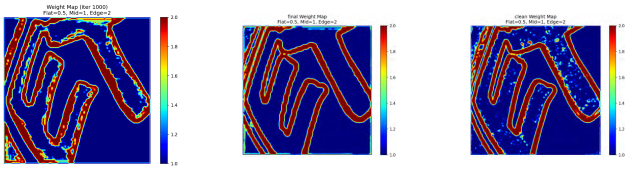
$$\mathcal{L}_{init} = \mathcal{L}_{rec} + \lambda \mathcal{L}_{cons}, \tag{5}$$

where

$$\mathcal{L}_{rec} = \left\| (\hat{\mathbf{x}} - \mathbf{y}) \odot \mathbf{M} \right\|_1, \tag{6}$$

$$\mathcal{L}_{cons} = \left\| (\hat{\mathbf{x}} - \hat{\mathbf{x}}_{base}) \odot \mathbf{M} \right\|_1. \tag{7}$$

Here, $\odot$ denotes element-wise multiplication, and $\|\cdot\|_1$ represents the $L_1$ norm. The hyperparameter $\lambda = 0.2$ balances the two loss terms. Normalization by $\|\mathbf{M}\|_1$ (the total number of masked pixels) ensures that the loss magnitude remains invariant to the mask ratio, thereby providing a consistent gradient signal throughout the training process. This stage imposes a uniform pixel-wise reconstruction constraint while preserving consistency between the multi-scale output and the base branch, thus providing a stable initialization for the subsequent optimization. As shown in Fig. 4, after the initial training stage, the spatial weight map $\mathbf{W}$ derived from the intermediate denoised estimate exhibits strong structural correspondence with the gradient field of the clean image. This alignment confirms that the estimated weight map can serve as a reliable fixed spatial prior in the following stage.



(a) Spatial weight map $\mathbf{W}$ (b) Gradient of denoised $\hat{\mathbf{x}}$ (c) Gradient of clean $\mathbf{x}$

Fig. 4: Validation of the edge-aware spatial weight map. The structural correspondence between the spatial weight map and the clean image gradients confirms that the estimation reliably identifies edges.

In the edge-aware optimization stage, the overall training objective is formulated as:

$$\mathcal{L}_{edge} = \mathcal{L}_{rec}^{edge} + \lambda \mathcal{L}_{cons}^{edge}, \tag{8}$$

where

$$\mathcal{L}_{rec}^{edge} = \left\| \mathbf{W} \odot (\hat{\mathbf{x}} - \mathbf{y}) \odot \mathbf{M} \right\|_1, \tag{9}$$

$$\mathcal{L}_{cons}^{edge} = \left\| \mathbf{W} \odot (\hat{\mathbf{x}} - \hat{\mathbf{x}}_{base}) \odot \mathbf{M} \right\|_1. \tag{10}$$

$\mathcal{L}_{rec}^{edge}$ emphasizes structurally informative regions via spatial weighting, while $\mathcal{L}_{cons}^{edge}$ mitigates scale-wise dominance by enforcing consistency between the fused multi-scale output and the original-resolution branch. This design enables balanced multi-scale learning while preserving fine structural details.

**Weight map construction.** A stabilized denoised estimate $\tilde{x}$ is obtained by averaging multiple forward passes with different random masks, thereby reducing estimation variance.

TABLE I: Quantitative comparisons (PSNR(dB) / SSIM) of different denoising methods on real datasets. The best results of the zero-shot category are **bolded**.

| Method | SIDD Val. | SIDD Ben. | FMDD |
|---|---|---|---|
| **self-supervised** | | | |
| Noise2Void [3] | 27.06/0.651 | 26.99/0.652 | - |
| NBR2NBR [5] | 27.94/0.604 | 27.90/0.679 | - |
| CVF-SID [9] | 34.81/0.944 | 34.71/0.917 | 32.73/0.843 |
| R2R [6] | 35.04/0.844 | 35.88/0.863 | - |
| APBSN [10] | 36.73/0.878 | 36.91/0.931 | 31.99/0.836 |
| LG-BPN [7] | 37.32/0.886 | 37.28/0.936 | - |
| ATBSN [11] | 37.88/0.946 | 37.78/0.944 | - |
| **zero-shot** | | | |
| DIP [14] | 32.11/0.740 | - | 32.90/0.854 |
| PD-denoising [23] | 33.97/0.820 | - | 33.01/0.856 |
| Self2Self [15] | 29.46/0.595 | 29.51/0.651 | 30.76/0.695 |
| APBSN-single [10] | 30.90/0.818 | 30.71/0.869 | 28.43/0.804 |
| ZS-N2N [17] | 25.59/0.565 | 30.19/0.428 | 31.65/0.767 |
| ScoreDVI [20] | 34.75/0.856 | 35.39/0.859 | 33.10/0.695 |
| TTT-MIM [24] | 33.22/0.638 | 34.91/0.805 | 31.41/0.819 |
| MASH [21] | 35.06/0.851 | 34.80/0.814 | 33.71/0.882 |
| Pixel2Pixel [18] | 30.29/- | 31.87/0.31 | 33.26/0.880 |
| BSN-INS [25] | 35.31/0.868 | 35.05/**0.922** | **33.95/0.885** |
| Ours | **36.87/0.880** | **36.59**/0.871 | 33.08/0.874 |

The estimate is then converted to grayscale $\tilde{x}_{gray}$ and its gradient magnitude is computed using Sobel operators:

$$G = \sqrt{(\nabla_h \tilde{x}_{gray})^2 + (\nabla_v \tilde{x}_{gray})^2}, \tag{11}$$

where $\nabla_h$ and $\nabla_v$ denote horizontal and vertical Sobel filters. The gradient magnitude is normalized to $[0, 1]$. Based on its distribution across the image, we compute two percentile thresholds corresponding to the 0.3 and 0.7 quantiles, denoted as $\tau_{low}$ and $\tau_{high}$ respectively. Pixels are then classified into three categories:

- Flat region: $G < \tau_{low}$
- Medium region: $\tau_{low} \leq G < \tau_{high}$
- Edge region: $G \geq \tau_{high}$

A spatial weight map $\mathbf{W} \in \mathbb{R}^{H \times W}$ is constructed by assigning fixed weights to each region: $w_{flat} = 0.5$, $w_{mid} = 1.0$, $w_{edge} = 2.0$.

## IV. Experiments

### A. Experiment Settings

**Datasets.** The proposed method is evaluated on the Smartphone Image Denoising Dataset (SIDD) [26] and the Fluorescence Microscopy Denoising Dataset (FMDD) [27]. We conduct ablation studies on the SIDD validation set and report results on the SIDD validation, the SIDD benchmark, and the FMDD dataset. For zero-shot methods, we adhere to the single-image adaptation protocol, where each test image is processed independently without any external training data.

**Implementation Details.** For the multi-scale architecture, we employ four DBSN branches at scales 1, 1/2, 1/4, and 1/8. Each DBSN branch adopts the same architecture as in [22], consisting of 9 dilated convolutional layers with a dilation pattern of [1,2,3,4,3,2,1] and 64 channels. Downsampling is performed via bilinear interpolation, and upsampling similarly uses bilinear interpolation to restore the original resolution. During training, we use the Adam optimizer with an initial learning rate of $1 \times 10^{-4}$ and a batch size of 1. The mask sampling probability $p$ in (1) is set to 0.3. The edge-aware loss configuration follows the description in Sec III-C. Training continues for a total of 20,000 iterations. For zero-shot single-image denoising, we apply the same training procedure independently to each test image.

TABLE II: Ablation Study on Multi-Scale Architecture.

| Scale 1/2 | Scale 1/4 | Scale 1/8 | Scale 1/16 | PSNR |
|---|---|---|---|---|
| × | × | × | × | 36.66 |
| ✓ | × | × | × | 36.75 |
| ✓ | ✓ | × | × | 36.83 |
| ✓ | ✓ | ✓ | × | **36.87** |
| ✓ | ✓ | ✓ | ✓ | 36.68 |

TABLE III: Ablation Study on Edge-Aware Spatial Loss.

| Edge-Aware Spatial Loss | PSNR | SSIM |
|---|---|---|
| w/o | 36.32 | 0.874 |
| w/ | 36.66 | 0.878 |

**Evaluation Metrics.** We adopt two widely used metrics for quantitative evaluation: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). Higher PSNR and SSIM values indicate better denoising performance in terms of pixel-wise fidelity and perceptual structure preservation.

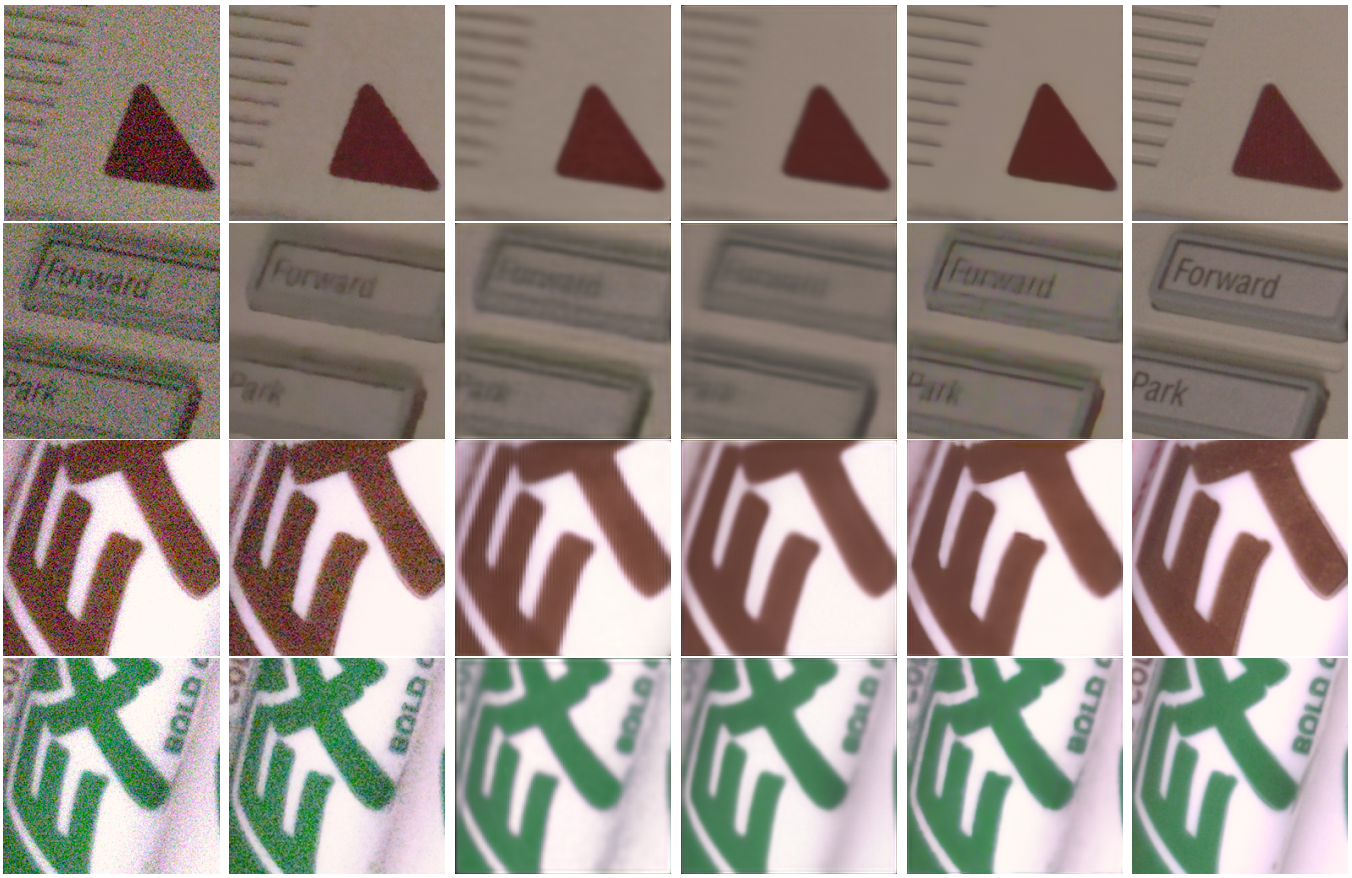
### B. Performance Evaluation

As shown in Table I, Our method achieves competitive performance among zero-shot approaches. Although EMBSD is trained exclusively on the test image itself, its performance is comparable to that of several self-supervised methods that rely on large-scale training datasets. Moreover, as illustrated in Fig. 5, our method demonstrates a clear advantage in preserving fine textural details.

The performance on FMDD is lower than that observed on SIDD. This discrepancy could be related to the intrinsic differences in signal characteristics between fluorescence microscopy images and natural images. First, although the low-resolution downsampling branch is designed to capture global context, the effective signals in FMDD occur at very fine scales and are prone to being lost during the downsampling process. When these degraded representations are subsequently upsampled and fused back into the original

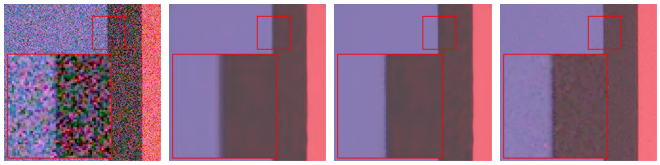
TABLE IV: Comparison of model parameters with different scale branch configurations.

| Scale 1 | Scale 1/2 | Scale 1/4 | Scale 1/8 | Params (M) |
|---|---|---|---|---|
| ✓ | × | × | × | 0.5 |
| ✓ | ✓ | × | × | 1.0 |
| ✓ | ✓ | ✓ | × | 1.5 |
| ✓ | ✓ | ✓ | ✓ | 2.0 |




(a) Noisy (b) ScoreDVI (c) MASH (d) BSN-INS (e) Ours (f) Clean

Fig. 5: Visual comparison of our method against other zero-shot denoising methods on SIDD validation images.



(a) Noisy (b) w/o Edge-Aware Spatial Loss (c) w/ Edge-Aware Spatial Loss (d) Clean

Fig. 6: Visual comparison of edge-aware spatial loss effect.

resolution, they tend to introduce a smoothing bias into the reconstructed image. Second, the extremely low signal-to-noise ratio of fluorescence microscopy images causes the gradient responses to become numerically entangled with noise fluctuations. Consequently, the edge-aware weight maps struggle to reliably distinguish weak anatomical edges, leading to residual artifacts and excessive smoothing of fine structural details.

### *C. Ablation Study*

To systematically analyze the contributions of each component in EMBSD, we conduct ablation studies on the SIDD validation set. We evaluate two key aspects: the multi-scale architecture and the edge-aware spatial loss weighting.

**Effect of Multi-Scale Architecture.** Table II shows the impact of adding different scale branches. The baseline single-scale model (only scale 1) achieves 36.66 dB PSNR. Adding the 1/2 scale branch improves PSNR to 36.75 dB, indicating that moderate downsampling helps capture mid-range contextual information. Further adding the 1/4 scale branch raises the PSNR to 36.83 dB, and including all four scales (1, 1/2, 1/4, 1/8) yields the best result of 36.87 dB. These results validate that explicit multi-scale modeling progressively enhances denoising performance by integrating both local details and global context. However, when an additional 1/16 scale branch is appended, the PSNR declines to 36.68 dB. This degradation can be attributed to the excessively coarse resolution at 1/16 scale, where the input size becomes too small for the blind-spot network to preserve meaningful spatial structures. Consequently, the upsampled output introduces smoothing artifacts and aliasing errors that outweigh the negligible gains in global receptive field, ultimately contaminating the fused multi-scale prediction. These results suggest that a four-scale configuration represents the optimal trade-off between contextual modeling capacity and spatial fidelity.

**Effect of Edge-Aware Spatial Loss.** Table III presents an ablation study comparing the uniform spatial loss with the proposed edge-aware loss. The *w/o* entry corresponds to the

uniform training objective $\mathcal{L}_{init}$ defined in (5), i.e., without the spatial weight map $\mathbf{W}$. Enabling the complete edge-aware spatial loss $\mathcal{L}_{edge}$ in (8) yields consistent improvements under the same single-scale backbone configuration, increasing PSNR from 36.32 dB to 36.66 dB (a gain of 0.34 dB) and SSIM from 0.874 to 0.878. Fig. 6 provides a visual comparison of this effect, demonstrating that the model trained with the edge-aware spatial loss restores sharper structural boundaries.

**Model Complexity Analysis.** To evaluate the parameter efficiency of our multi-scale design, we compare parameter counts across different configurations in Table IV. The baseline single-scale model is compact with 0.5M parameters. Each additional DBSN branch (at 1/2, 1/4, and 1/8 scales) introduces an extra 0.5M parameters.

## V. Conclusion

In summary, we propose EMBSD, an explicit multi-scale blind-spot denoising framework for real-world images. It employs multi-resolution dilated blind-spot networks to capture cross-scale noise and introduces an edge-aware spatial loss to balance detail preservation and noise suppression. By combining multi-scale modeling with spatially adaptive optimization, EMBSD effectively disentangles noise and structure. Experiments demonstrate competitive or superior performance among zero-shot methods with improved structural fidelity.

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